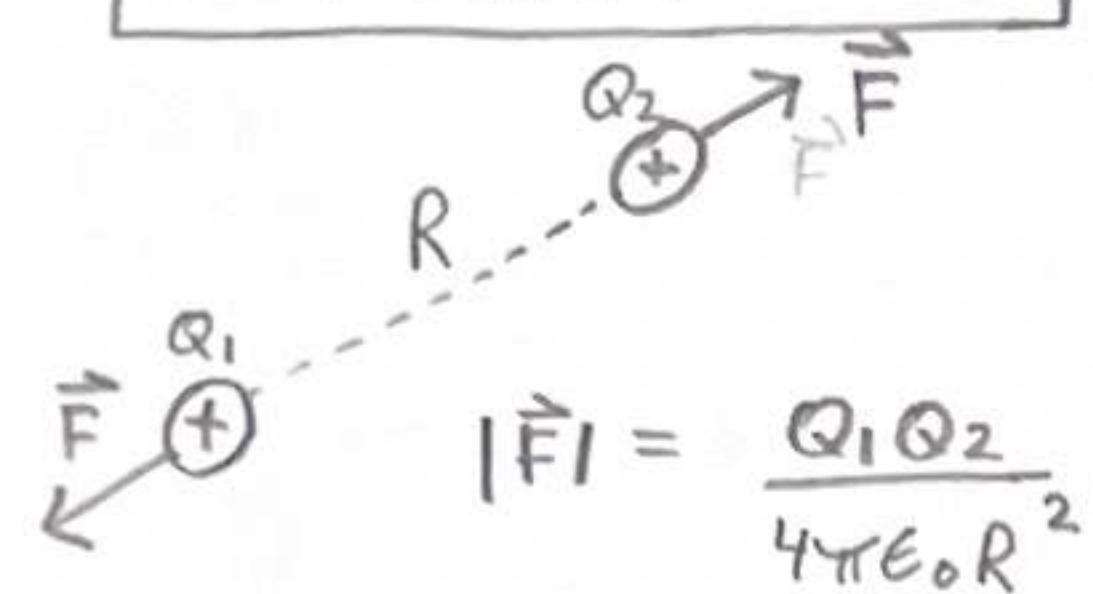


ELEC 211 Math 264

Coulomb's Law



Work

"E field exerts a force on charged particles"

$$F = QE$$

"To move this particle we need to exert a force, move in direction $\hat{a}_L$ "

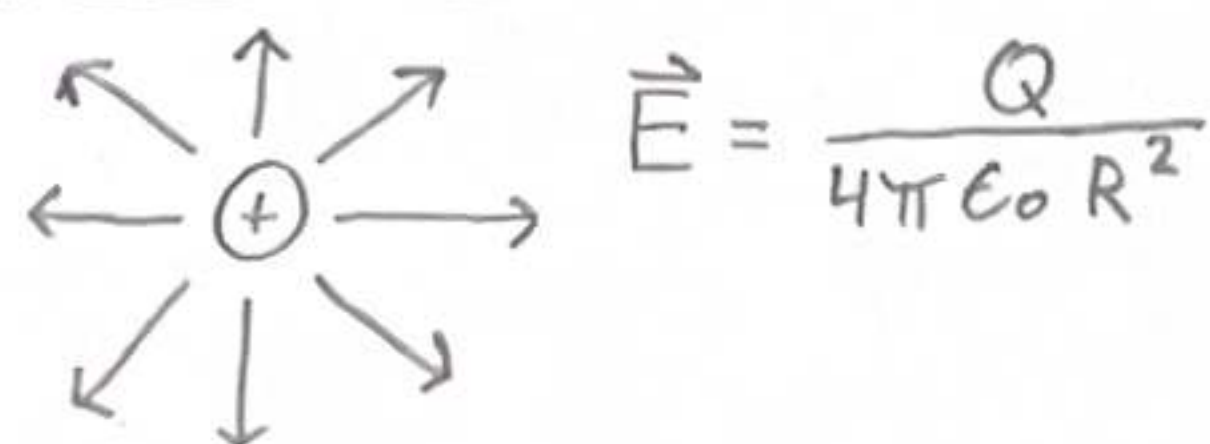
$$F = -Q\vec{E} \cdot \hat{a}_L$$

Move q towards Q

$$W = \frac{qQ}{4\pi\epsilon_0} \left[\frac{1}{a} - \frac{1}{b} \right]$$

Electric Fields

Point Charge



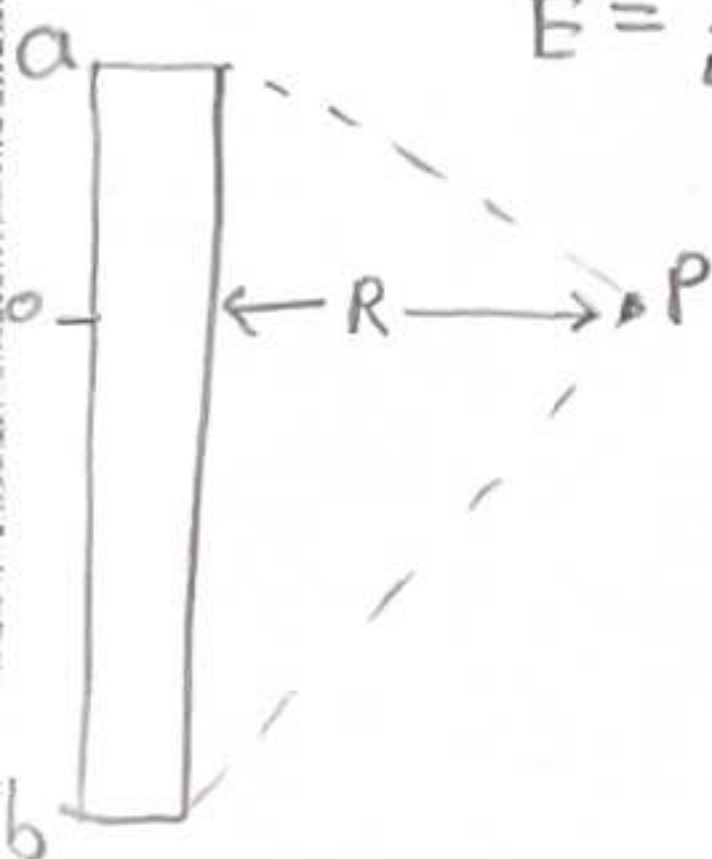
Line of Charge

Infinite:

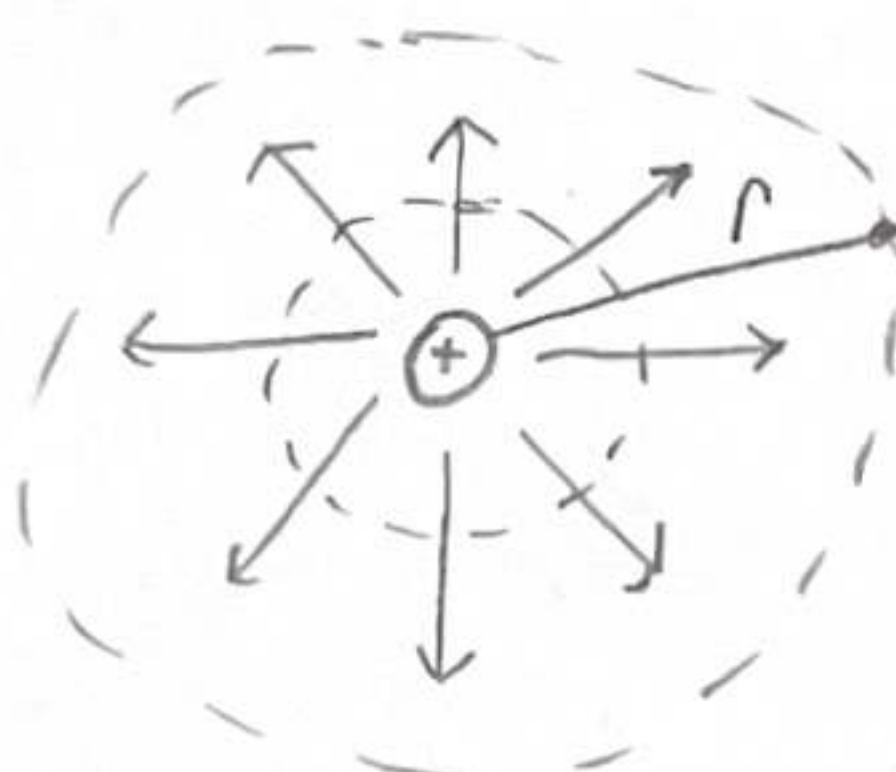
$$\vec{E} = \frac{\rho_L}{2\pi\epsilon_0 R}$$

Finite:

$$\vec{E} = \frac{\rho_L}{4\pi\epsilon_0 R} \left[\frac{b}{\sqrt{R^2 + b^2}} + \frac{a}{\sqrt{R^2 + a^2}} \right]$$



Electric Potential



$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r} \text{ (scalar)}$$

"Constant around equipotential sphere"

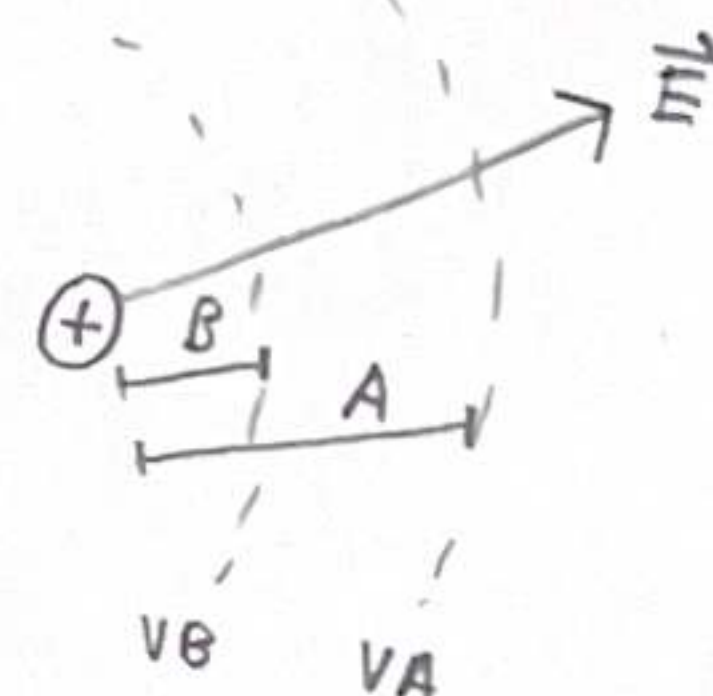
Electric field from Potential

$$\vec{E} = -\nabla V$$

$$\vec{E} = -\left\langle \hat{i} \frac{d(V)}{dx}, \hat{j} \frac{d(V)}{dy}, \hat{k} \frac{d(V)}{dz} \right\rangle$$

Potential from Electric Field

$$\Delta V = V_A - V_B = -\int_B^A \vec{E} \cdot d\vec{s}$$



Sheet of Charge (Infinite)



Finding Potential from Conservative Vector Field

*If F is conservative, there exists f where $\nabla f = \langle f_x, f_y, f_z \rangle = \vec{F}$

Ex: $\vec{F} = \langle 2x+yz, xz, yx \rangle$

$$\langle f_x, f_y, f_z \rangle = \langle 2x+yz, xz, yx \rangle$$

$$f_x = 2x+yz \rightarrow \int (2x+yz) dx$$

$$f_y = xz$$

$$f_z = yx$$

$$= x^2 + xy + g(y, z)$$

$$f = x^2 + xy + g(y, z)$$

$$f_y = xz + g_y(y, z)$$

$$\therefore g_y(y, z) = 0$$

$$\int 0 dy = h(z)$$

$$\therefore g(y, z) = h(z)$$

$$f = x^2 + xy + h(z)$$

$$f_z = xy + h'(z)$$

$$\therefore h'(z) = 0$$

$$\int 0 dz = C$$

$$\therefore h(z) = C$$

$$\therefore f = x^2 + xy + C$$

$$\nabla f = \langle 2x+yz, xz, xy \rangle = \vec{F}$$

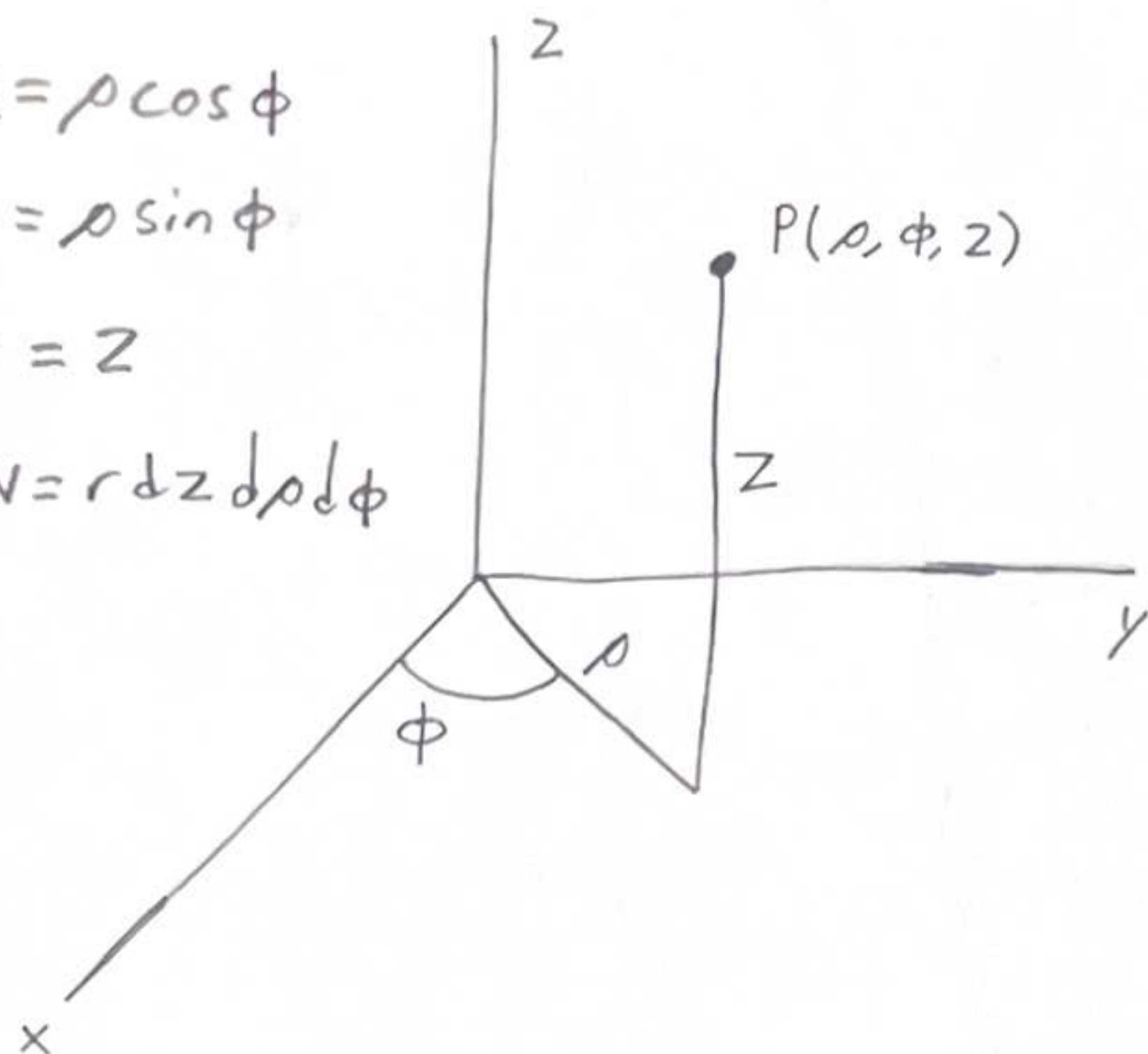
Cylindrical Coordinate System

$$x = \rho \cos \phi$$

$$y = \rho \sin \phi$$

$$z = z$$

$$dV = r dr d\phi dz$$



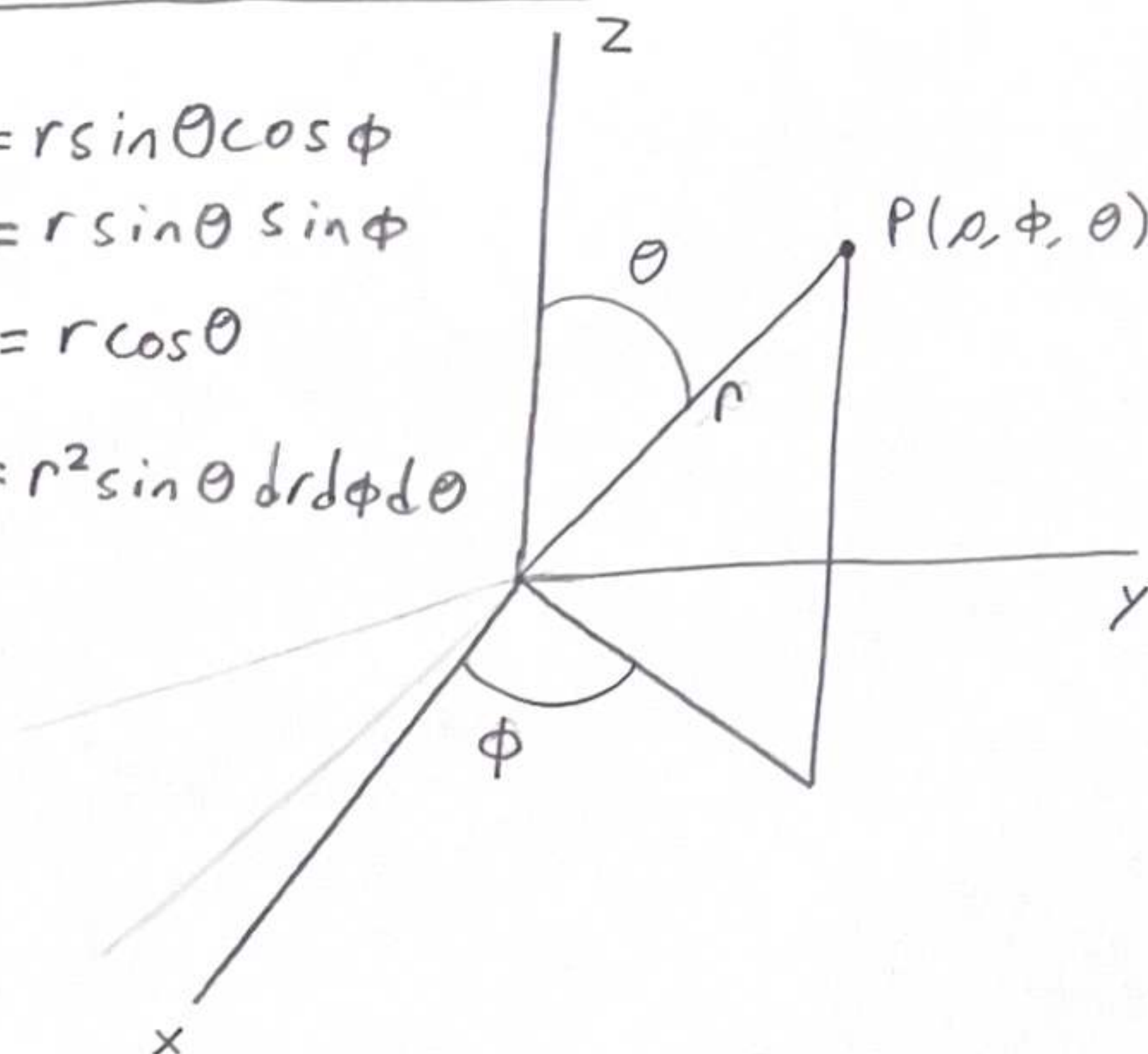
Spherical Coordinate System

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$dV = r^2 \sin \theta dr d\phi d\theta$$



Electric Flux

$$\Psi = Q_{enc}$$

Flux Density:

$$|\vec{D}| = \frac{\Psi}{\text{area}}$$

$$\vec{D} = \epsilon \vec{E}$$

$$\Psi = \oint_S \vec{D} \cdot d\vec{s}$$

where $d\vec{s}$ is $(\hat{n})$ to surface

if we have a surface parameterized by $r(u, v)$

$$d\vec{s} = \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v}$$

Line Integrals

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt$$

where the curve is given by

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k} \quad a \leq t \leq b$$

Fundamental Theorem: (Conservative)

if C , given by $\vec{r}(t)$, is smooth, and $\vec{F}$ has potential function f such that $\nabla f = \vec{F}$.

$$\int_C \nabla f \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$$

Gauss's Divergence Theorem

$$\oint_S \vec{D} \cdot \hat{n} dS = Q_{enc}$$

All equivalent

$$\iiint_R (\nabla \cdot \vec{D}) dv = \iiint_R \rho dv$$

Parameterizations:

Ellipse

$$\rightarrow x = a \cos(t) \quad y = b \sin(t) \quad 0 \leq t \leq 2\pi$$

Intersection of Functions

$$y = f(x) \quad z = g(x) \rightarrow \begin{matrix} x = t \\ y = f(t) \\ z = g(t) \end{matrix} \quad \text{choose bounds}$$

Line Segment

$$(x_0, y_0, z_0) \text{ to } (x_1, y_1, z_1)$$

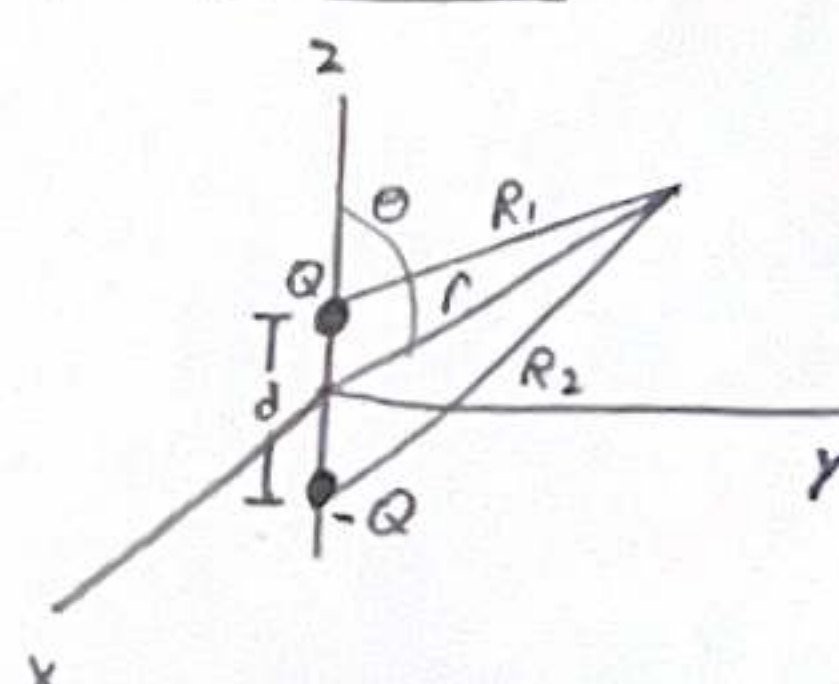
$$\vec{r}(t) = (1-t)(x_0, y_0, z_0) + t(x_1, y_1, z_1) \quad 0 \leq t \leq 1$$

Surface Integrals

Surface given by $r(u, v)$

$$\iint_S f(x, y, z) dS = \iint_D f(\vec{r}(u, v)) \|\vec{r}_u \times \vec{r}_v\| du dv$$

Dipoles



$$V = \frac{Qd \cos \theta}{4\pi \epsilon_0 r^2}$$

$$\vec{E} = -\nabla V$$

Dipole Moment Vector

$$\text{let } p = Qd$$

$$d \cdot \hat{a}_r = d \cos \theta$$

$$V = \frac{p \cdot \hat{a}_r}{4\pi \epsilon_0 r^2}$$

$$V = \frac{1}{4\pi \epsilon_0 |r - r'|^2} p \cdot \frac{r - r'}{|r - r'|}$$

Current

$$I = \frac{dQ}{dt}$$

(Current Density):

$$J = (A/m^2)$$

$$J = \sigma E \quad \text{conductivity}$$

Point Form Ohm's Law

*if parallel

$$V = \frac{L}{\sigma S} \cdot I = IR$$

$$J = \rho_v \vec{V}$$

Resistance

$$R = \frac{L}{\sigma S} = \rho \frac{L}{S}$$

$$\text{where } \rho = \frac{1}{\sigma}$$

$$I = \iint_S \vec{J} \cdot d\vec{s}$$

general case:

$$R = \frac{V_{ab}}{I} = \frac{-\int_a^b \vec{E} \cdot d\vec{l}}{\iint_S \vec{J} \cdot d\vec{s}}$$

Polarization

$$P = \frac{\vec{p}}{V} \quad \text{where: } \vec{p} = q\vec{d}$$

$V = \text{volume}$

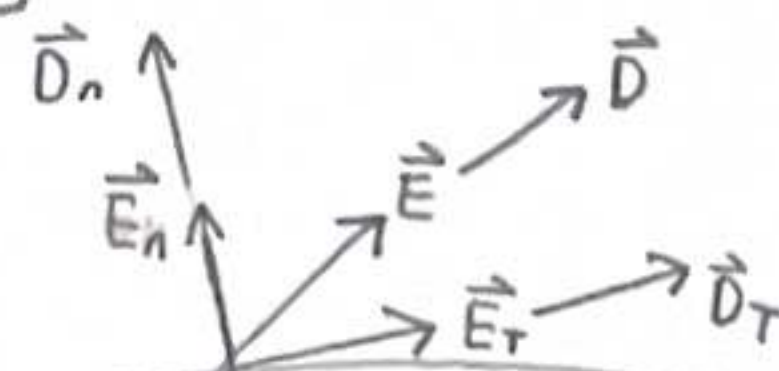
Boundary Conditions

* Use to find the field on one side of the boundary if we know the field on the other side

Cases:

- Dielectric (ϵ_1) and Dielectric (ϵ_2)
- Conductor and Dielectric
- Conductor and free space

Case 3: (Conductor and Free Space)



Conductor ($\vec{E} = 0$)

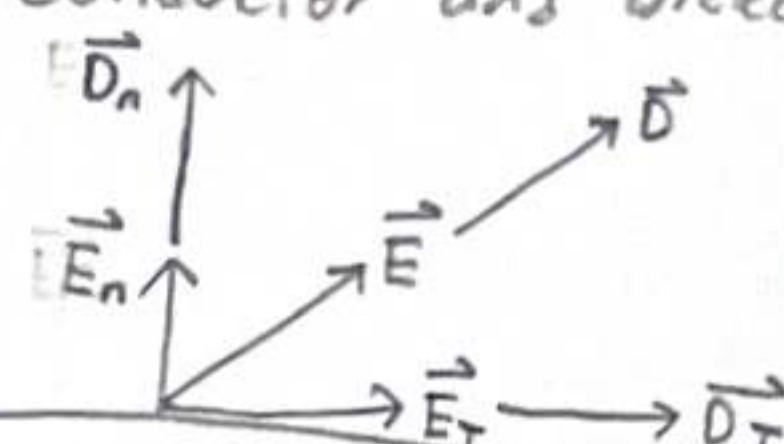
equipotential surface

Characteristics:

- $E_t = D_t = 0$
- $\epsilon_0 E_n = D_n = \rho_s$

* The electric field is external to the conductor and normal to its surface

Case 2: (Conductor and Dielectric)



Conductor ($\vec{E} = 0$)

Characteristics:

- $E_t = D_t = 0$
- $\epsilon_0 \epsilon_r E = D_n = \rho_s$

Case 1: (Dielectric and Dielectric)

②

$$\epsilon_2 = \epsilon_0 \epsilon_{r2}$$

Characteristics:

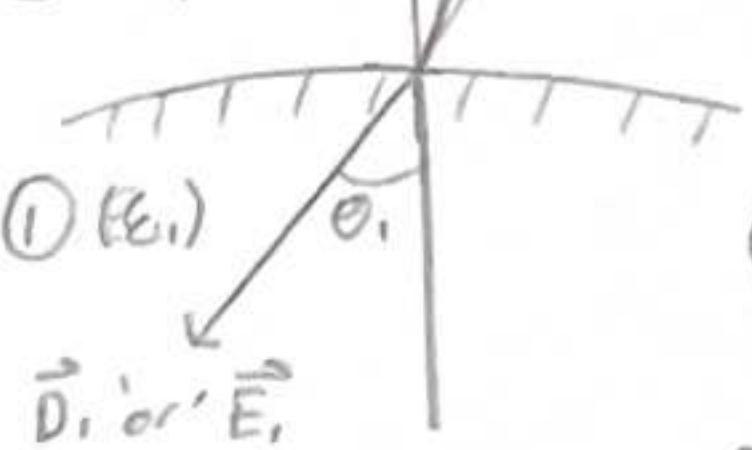
- $E_{1T} = E_{2T}$
- $\frac{D_{1T}}{\epsilon_1} = \frac{D_{2T}}{\epsilon_2}$
- $D_{2n} - D_{1n} = \rho_s$
- * if $\rho_s = 0$ (@ Boundary)
- $D_{2n} = D_{1n}$
- $\epsilon_1 E_{1n} = \epsilon_2 E_{2n}$

①

$$\epsilon_1 = \epsilon_0 \epsilon_{r1}$$

Refraction

② (ϵ_2)



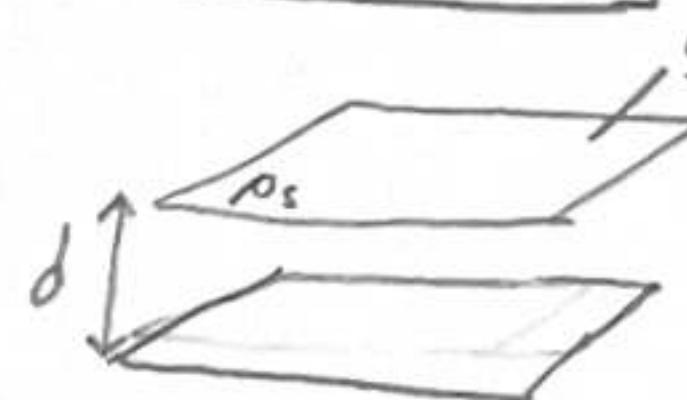
① (ϵ_1)



Characteristics:

- $E_{1T} = E_{2T}$
- $D_{1n} = D_{2n}$
- $E_1 \sin \theta_1 = E_2 \sin \theta_2$
- $D_1 \cos \theta_1 = D_2 \cos \theta_2$
- $\epsilon_1 E_1 \cos \theta_1 = \epsilon_2 E_2 \cos \theta_2$
- $\frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_1}{\epsilon_2}$

Capacitors

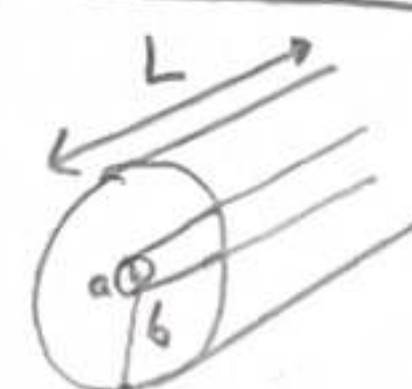


$$C = \frac{Q}{V}$$

$$C = \frac{\epsilon_0 S}{d}$$

(Parallel-Plate, constant ρ_s)

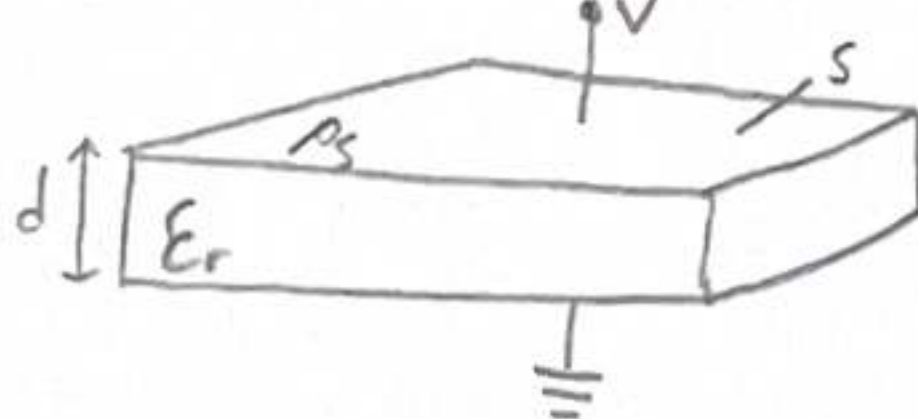
Coaxial Cable



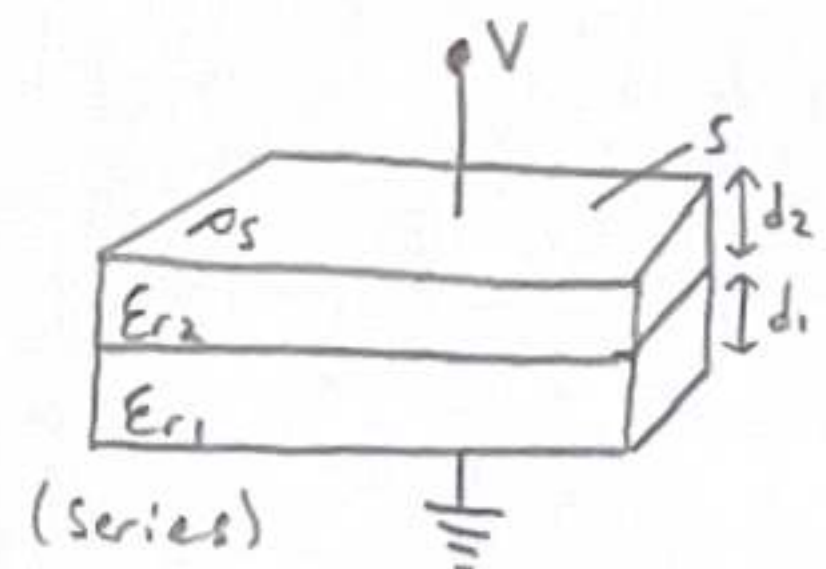
$$C = \frac{Q}{V}$$

$$C = \frac{2\pi\epsilon_0 L}{\ln(b/a)}$$

Capacitor with Dielectrics



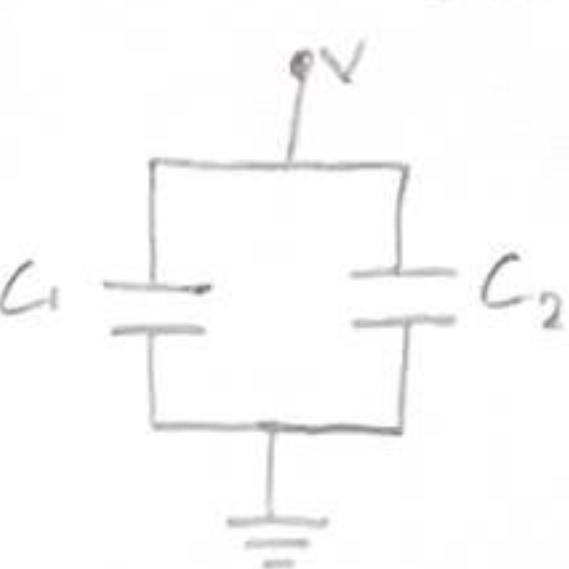
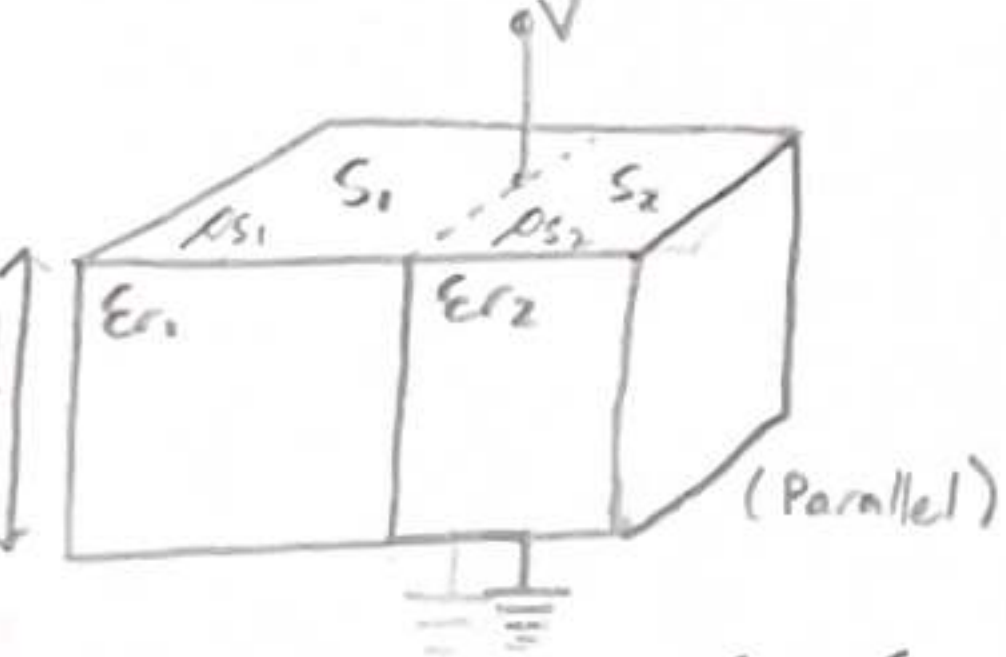
$$C = \frac{\epsilon_0 \epsilon_r S}{d}$$



$$C = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2}} \quad C_1 = \frac{\epsilon_1 \epsilon_0 S}{d_1} \quad C_2 = \frac{\epsilon_2 \epsilon_0 S}{d_2}$$

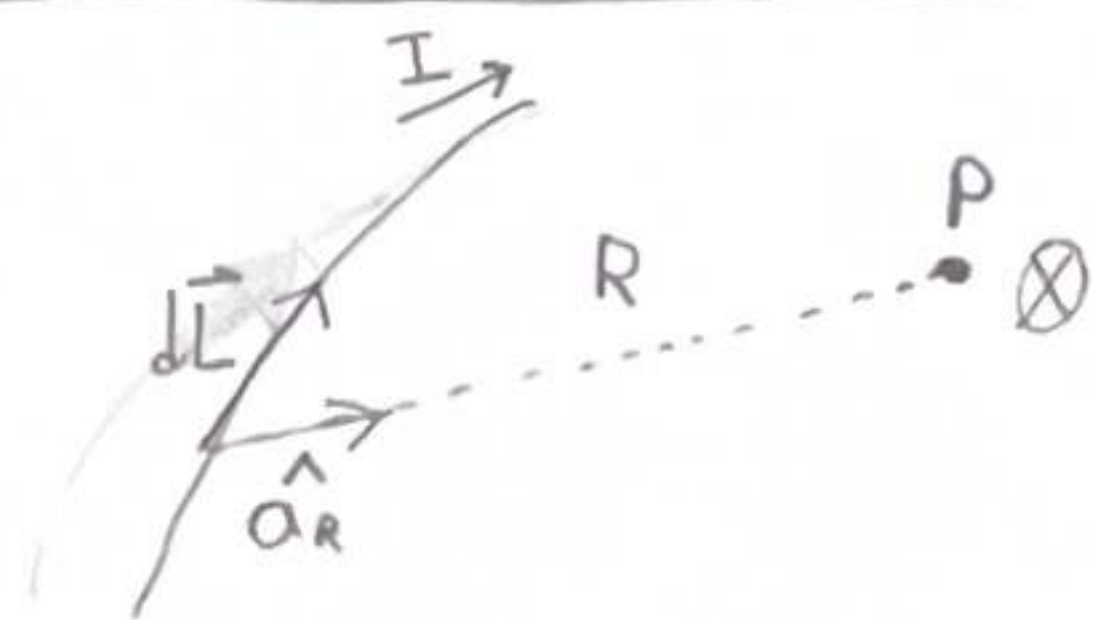
Graded Dielectric

$$C = \frac{\epsilon_1 \epsilon_2 \epsilon_0 S}{\epsilon_2 d_1 + \epsilon_1 d_2}$$



$$C = \frac{\epsilon_1 \epsilon_0 S_1}{d} + \frac{\epsilon_2 \epsilon_0 S_2}{d} = \frac{\epsilon_0}{d} (\epsilon_1 S_1 + \epsilon_2 S_2)$$

Biot-Savart Law



$$d\vec{H} = \frac{I d\vec{L} \times \hat{a}_R}{4\pi R^2}$$

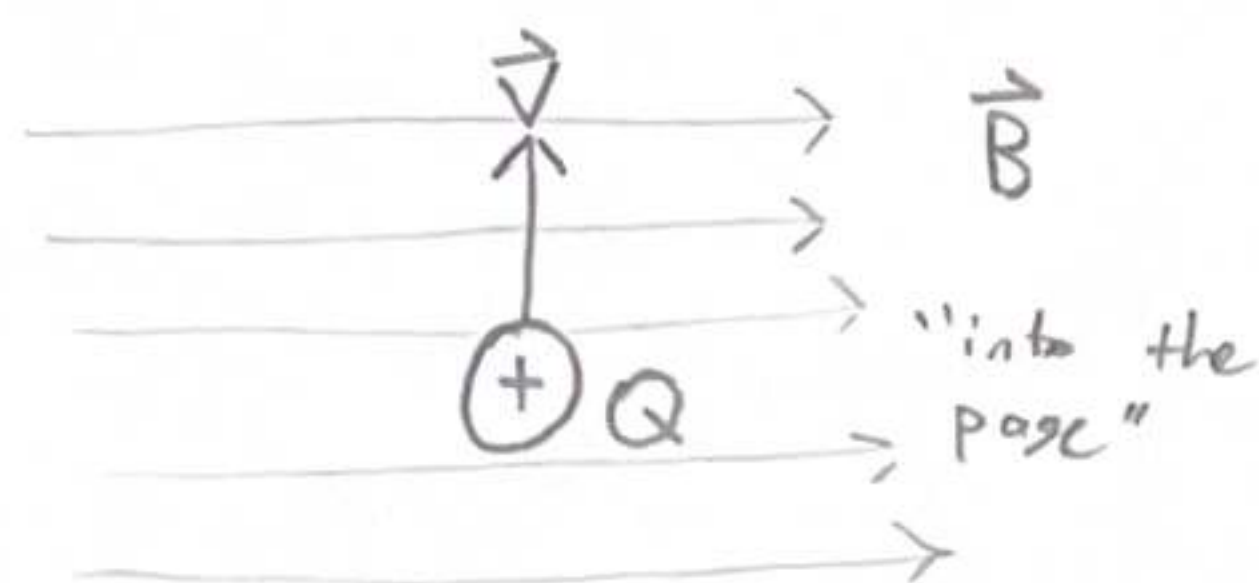
Divergence Theorem

$$\vec{H} = \oint \frac{I d\vec{L} \times \hat{a}_R}{4\pi R^2}$$

$$\Phi = \oint_S \vec{B} \cdot d\vec{S} = 0$$

$$\therefore \nabla \cdot \vec{B} = 0$$

Force on a Moving Charge



$$\vec{F} = Q(\vec{E} + \vec{v} \times \vec{B})$$

Surface Current

$$\vec{H} = \iint_S \frac{\vec{k} \times \hat{a}_R dS}{4\pi R^2}$$

Volume Current

$$\vec{H} = \iiint_{vol} \frac{\vec{J} \times \hat{a}_R dV}{4\pi R^2}$$

Magnetic Flux Density

$$\vec{B} = \mu_0 \vec{H} \text{ where:}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

$$\Phi = \iint_S \vec{B} \cdot d\vec{S}$$

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PHYSICAL CONSTANTS

Permittivity of free space:	$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$	Permeability of free space:	$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$
Electron charge:	$e = 1.602 \times 10^{-19} \text{ C}$	Electron mass:	$m = 9.109 \times 10^{-31} \text{ kg}$
Speed of light in vacuum:	$c = 2.998 \times 10^8 \text{ m/s}$		

ELECTROSTATIC PRINCIPLES

Coulomb's Law:	$\mathbf{F}_2 = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{R_{12}^2} \mathbf{a}_{12}$	$\mathbf{a}_{12} = \frac{\mathbf{R}_{12}}{ \mathbf{R}_{12} } = \frac{\mathbf{R}_{12}}{R_{12}} = \frac{\mathbf{r}_2 - \mathbf{r}_1}{ \mathbf{r}_2 - \mathbf{r}_1 }$
Point Charge Q at O :	$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \mathbf{a}_r, V = \frac{Q}{4\pi\epsilon_0 r}$	(r comes from spherical coords)
Line Charge, density ρ_L , on z -axis:	$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0} \left(\frac{\mathbf{a}_\rho}{\rho} \right), V = \frac{\rho_L}{2\pi\epsilon_0} \ln\left(\frac{1}{\rho}\right)$	(ρ comes from cylindrical coords)
Sheet Charge, density ρ_S , on $z = 0$:	$\mathbf{E} = \pm \frac{\rho_S}{2\epsilon_0} \mathbf{a}_z, V = -\frac{\rho_S z }{2\epsilon_0}$	(Both ρ_S and ρ_L must be constant here.)
Electric Flux Density:	$(\epsilon/\epsilon_0) \mathbf{D} = \epsilon \mathbf{E}$	($\epsilon = \epsilon_0 \epsilon_r$ in general; $\epsilon_r = 1$ in free space)
Gauss's Law, I:	$(C) Q_{\text{enc}} = \Psi$, where	$\Psi = \oiint_S \mathbf{D} \cdot \hat{\mathbf{n}} dS$ is net outward flux
Gauss's Law, II:	$Q_{\text{enc}} = \iiint_V \rho_v dv$, where	$\rho_v = \nabla \cdot \mathbf{D}$ gives charge density
Electric field and potential:	$(V/m) \mathbf{E} = -\nabla V$	$V(B) - V(A) = - \int_A^B \mathbf{E} \cdot d\mathbf{L}$ (path indep)
Generalized Poisson Equation:	$\nabla \cdot (\epsilon \nabla V) = -\rho_v$	(Case $\rho_v = 0, \epsilon = \text{const}$ is Laplace's Equation.)
Energy in Electrostatic Field:	$W_E = \frac{1}{2} \iiint_V \mathbf{D} \cdot \mathbf{E} dv = \frac{1}{2} \iiint_V \epsilon \mathbf{E} ^2 dv$	

CONDUCTORS, CURRENT, RESISTANCE

Ideal conductor (" $\sigma \rightarrow \infty$ "):	$\mathbf{E}_T = 0$	$V = \text{const.}$
Ideal conductor boundary:	$\mathbf{E} \parallel \hat{\mathbf{n}}$	$\rho_S = \mathbf{D} \cdot \hat{\mathbf{n}}$
Current and conductivity:	$\mathbf{J} = \sigma \mathbf{E}$ "Ohm's Law I"	$I = \iint_S \mathbf{J} \cdot \hat{\mathbf{n}} dS$
	$\mathbf{J} = \rho_v \mathbf{v}$	$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$
Simple Resistor (length L , constant cross-section S , constant conductivity σ):		$R = \frac{L}{\sigma S}$
Fancy Resistor (all current from A to B crosses surface S "Ohm's Law II"):		$R = \frac{ \Delta V }{ I } = \frac{\left - \int_A^B \mathbf{E} \cdot d\mathbf{L} \right }{\left \iint_S \mathbf{J} \cdot \hat{\mathbf{n}} dS \right }$

CAPACITORS AND DIELECTRICS

Permittivity:	$\epsilon = \epsilon_r \epsilon_0$	<u>Coaxial:</u>	(Gauss's Law still works, as above)
Polarization:	$\mathbf{P} = \mathbf{D} - \epsilon_0 \mathbf{E}$	$\frac{C}{L} = \frac{2\pi\epsilon}{\ln(b/a)}$	
Simple Capacitor (parallel plates of area S , separation d):			$C = \frac{\epsilon S}{d}$ stores $W_E = \frac{1}{2} CV^2$ Joules
Fancy Capacitor (surface S is one plate; points A, B on opposite plates):			$C = \frac{ Q }{ \Delta V } = \frac{\left \iint_S \mathbf{D} \cdot \hat{\mathbf{n}} dS \right }{\left - \int_A^B \mathbf{E} \cdot d\mathbf{L} \right }$
Dielectric interface with normal $\mathbf{n}$:	$\mathbf{D}_1 \cdot \mathbf{n} = \mathbf{D}_2 \cdot \mathbf{n}$ AND $D_{1N} = D_{2N}$		$\mathbf{E}_1 \times \mathbf{n} = \mathbf{E}_2 \times \mathbf{n}$ $E_{1T} = E_{2T}$
		$\frac{\tan(\theta_1)}{\tan(\theta_2)} = \frac{\epsilon_1}{\epsilon_2}$	

FPL: $\Phi = \int_{z=0}^L \int_{\rho=0}^b B \, d\rho \, dz$

MAGNETOSTATICS

Biot-Savart Law:

$$d\mathbf{H} = \frac{I d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$$

$$\mathbf{H} = \int \frac{I d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$$

Current I flowing in filament $\rho = 0$, direction $\mathbf{a}_z$:

$$\mathbf{H} = \frac{I}{2\pi\rho} \mathbf{a}_\phi; \text{ or, for segment, } \mathbf{H} = \frac{I}{4\pi\rho} (\sin\alpha_2 - \sin\alpha_1) \mathbf{a}_\phi$$

Current sheet with density $\mathbf{K}$ [A/m], normal $\hat{\mathbf{n}}$:

$$(\text{A/m}) \quad \mathbf{H} = \frac{1}{2} \mathbf{K} \times \hat{\mathbf{n}}$$

$$I = \int \mathbf{K} \cdot d\mathbf{w}$$

Current crossing surface S , from current density $\mathbf{J}$:

$$I = \iint_S \mathbf{J} \cdot d\mathbf{S}$$

$$\mathbf{J} = \nabla \times \mathbf{H}$$

Ampère's Circuital Law (ACL):

$$I = \oint \mathbf{H} \cdot d\mathbf{L}$$

(compare Stokes's Theorem)

Magnetic Flux Density:

$$\left[\frac{\text{Wb}}{\text{m}^2} \right] \quad \mathbf{B} = \mu \mathbf{H}$$

$$\mu = \mu_r \mu_0$$

Magnetic Flux (Wb):

$$(\text{Wb}) \quad \Phi = \iint_S \mathbf{B} \cdot d\mathbf{S}$$

$$\oiint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

Energy in Steady Magnetic Field:

$$W_H = \frac{1}{2} \iiint_{\mathcal{R}} \mathbf{B} \cdot \mathbf{H} \, dv = \frac{1}{2} \iiint_{\mathcal{R}} \mu |\mathbf{H}|^2 \, dv$$

Magnetic Force on Moving Charge:

$$(\text{N}) \quad \mathbf{F} = Q\mathbf{v} \times \mathbf{B}$$

$$\mathbf{J} = \mathbf{v} \rho_v$$

Magnetic Force on Current Filament:

$$d\mathbf{F} = I d\mathbf{L} \times \mathbf{B}$$

$$\mathbf{F} = \int_C I d\mathbf{L} \times \mathbf{B} = - \int_C I \mathbf{B} \times d\mathbf{L}$$

Magnetic Force on Current Sheet or Cloud:

$$d\mathbf{F} = (\mathbf{K} dS) \times \mathbf{B}$$

$$d\mathbf{F} = (\mathbf{J} dv) \times \mathbf{B}$$

Magnetic Dipole Moment ($\mathbf{m} = \mathbf{p}_m$):

$$d\mathbf{m} = I d\mathbf{S}$$

$$\mathbf{m} = NIS\hat{\mathbf{n}}$$

Magnetic Torque on Given Dipole:

$$\vec{\tau} = \mathbf{m} \times \mathbf{B}$$

$$|\vec{\tau}| = NI |\mathbf{B}| |\mathbf{S}|, \text{ if } \mathbf{B} \perp \mathbf{S}$$

Review: Force $\mathbf{F}$ with moment arm $\mathbf{R}$ gives torque:

$$\vec{\tau} = \mathbf{R} \times \mathbf{F}$$

INDUCTORS AND MAGNETIC MATERIALS

Permeability:

$$\mu = \mu_r \mu_0$$

Simple inductor (N filaments, current I in each):

$$L = \frac{N\Phi}{I}$$

$$\text{stores } W_H = \frac{1}{2} LI^2 \text{ Joules}$$

Mutual Inductance:

$$M_{12} = \frac{N_2 \Phi_{12}}{I_1} = \frac{N_1 \Phi_{21}}{I_2} = M_{21}$$

Material interface with normal $\mathbf{n}$:

$$\mathbf{B}_1 \cdot \mathbf{n} = \mathbf{B}_2 \cdot \mathbf{n}$$

$$\mathbf{H}_1 \times \mathbf{n} = \mathbf{H}_2 \times \mathbf{n}$$

MAGNETIC CIRCUITS

Magnetomotive force (simple setup N turns, current I): $V_m = NI$

Magnetomotive force (general filament from A to B):

$$V_m(B) - V_m(A) = - \int_A^B \mathbf{H} \cdot d\mathbf{L} \quad (\text{path restrictions apply})$$

Reluctance (cross-section S , length ℓ):

$$\mathcal{R} = \frac{V_m}{\Phi} = \frac{\ell}{\mu S} \quad (\text{integral defining } \Phi \text{ shown above})$$

Air-gap force (cross-section S):

$$\mathbf{F} = \frac{1}{2\mu_0} |\mathbf{B}|^2 S \hat{\mathbf{n}}$$

MAXWELL'S EQUATIONS (POINT FORM, GENERAL CASE set $\frac{\partial \mathbf{B}}{\partial t} = 0$ and $\frac{\partial \mathbf{D}}{\partial t} = 0$ in static situations)

$$\nabla \cdot \mathbf{D} = \rho_v$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

TIME-VARYING FIELDS

Faraday's Law (case of $N = 1$ current filament):

$$\text{cmf} = - \frac{d\Phi}{dt} = - \iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot \hat{\mathbf{n}} \, dS \quad (\text{units: Volts})$$

$$\text{cmf} = \oint_C \mathbf{E} \cdot d\mathbf{L} \quad (\text{loop shape matters!})$$

$$\text{emf} = BLV \sin \theta$$

VECTOR IDENTITIES

For $\mathbf{u} = u_x \mathbf{a}_x + u_y \mathbf{a}_y + u_z \mathbf{a}_z$, $\mathbf{v} = v_x \mathbf{a}_x + v_y \mathbf{a}_y + v_z \mathbf{a}_z$, $\mathbf{w} = w_x \mathbf{a}_x + w_y \mathbf{a}_y + w_z \mathbf{a}_z$,

$$\mathbf{u} \cdot \mathbf{v} = u_x v_x + u_y v_y + u_z v_z = |\mathbf{u}| |\mathbf{v}| \cos(\theta), \quad 0 \leq \theta \leq \pi$$

$$|\mathbf{u}| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = \sqrt{u_x^2 + u_y^2 + u_z^2}$$

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix} = \langle u_y v_z - u_z v_y, u_z v_x - u_x v_z, u_x v_y - u_y v_x \rangle$$

$$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}| |\mathbf{v}| \sin \theta$$

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \mathbf{v} \cdot (\mathbf{w} \times \mathbf{u}) = \mathbf{w} \cdot (\mathbf{u} \times \mathbf{v})$$

$$\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w}) \mathbf{v} - (\mathbf{u} \cdot \mathbf{v}) \mathbf{w}$$

DISTANCES AND PROJECTIONS

From point (x_0, y_0, z_0) to plane $Ax + By + Cz = D$:

$$s = \frac{|Ax_0 + By_0 + Cz_0 - D|}{\sqrt{A^2 + B^2 + C^2}}$$

$$\mathbf{F} = \text{proj}_{\mathbf{u}}(\mathbf{F}) + \text{orth}_{\mathbf{u}}(\mathbf{F})$$

$$\text{proj}_{\mathbf{u}}(\mathbf{F}) = \left(\frac{\mathbf{F} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \right) \mathbf{u}$$

DERIVATIVE IDENTITIES valid for smooth scalar-valued ϕ, ψ and smooth vector-valued $\mathbf{F}, \mathbf{G}$

$$\nabla(\phi\psi) = \phi \nabla\psi + \psi \nabla\phi$$

$$\nabla \cdot (\mathbf{F} \times \mathbf{G}) = (\nabla \times \mathbf{F}) \cdot \mathbf{G} - \mathbf{F} \cdot (\nabla \times \mathbf{G})$$

$$\nabla \cdot (\phi \mathbf{F}) = (\nabla \phi) \cdot \mathbf{F} + \phi (\nabla \cdot \mathbf{F})$$

$$\nabla \times (\mathbf{F} \times \mathbf{G}) = \mathbf{F}(\nabla \cdot \mathbf{G}) - \mathbf{G}(\nabla \cdot \mathbf{F}) - (\mathbf{F} \cdot \nabla) \mathbf{G} + (\mathbf{G} \cdot \nabla) \mathbf{F}$$

$$\nabla \times (\phi \mathbf{F}) = (\nabla \phi) \times \mathbf{F} + \phi (\nabla \times \mathbf{F})$$

$$\nabla(\mathbf{F} \cdot \mathbf{G}) = \mathbf{F} \times (\nabla \times \mathbf{G}) + \mathbf{G} \times (\nabla \times \mathbf{F}) + (\mathbf{F} \cdot \nabla) \mathbf{G} + (\mathbf{G} \cdot \nabla) \mathbf{F}$$

$$\nabla \times (\nabla \phi) = \mathbf{0} \quad (\text{curl grad} = 0)$$

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0 \quad (\text{div curl} = 0)$$

$$\nabla^2 \phi(x, y, z) = \nabla \cdot \nabla \phi(x, y, z) = \text{div grad } \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$$

SURFACE NORMALS AND AREA ELEMENTS

For any oriented surface normal $\mathbf{n} \neq \mathbf{0}$, $d\mathbf{S} = \hat{\mathbf{n}} dS = \frac{\mathbf{n}}{|\mathbf{n} \cdot \mathbf{a}_x|} dx dy = \frac{\mathbf{n}}{|\mathbf{n} \cdot \mathbf{a}_y|} dx dz = \frac{\mathbf{n}}{|\mathbf{n} \cdot \mathbf{a}_z|} dy dz, \quad dS = |d\mathbf{S}|$

Graph Surface $z = f(x, y)$: normal $\mathbf{n} = \pm \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, -1 \right\rangle \quad \hat{\mathbf{n}} dS = \pm \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, -1 \right\rangle dx dy$

Level Surface $G(x, y, z) = 0$: normal $\mathbf{n} = \pm \nabla G(x, y, z) \quad (\text{choose sign to orient})$

Parametric Surface $\langle x, y, z \rangle = \mathbf{R}(u, v)$: $d\mathbf{S} = \pm \left(\frac{\partial \mathbf{R}}{\partial u} \times \frac{\partial \mathbf{R}}{\partial v} \right) du dv \quad (\text{choose sign to orient; } \hat{\mathbf{n}} = \frac{d\mathbf{S}}{|d\mathbf{S}|})$

CARTESIAN COORDINATES (x, y, z)

Line Element: $d\mathbf{L} = \mathbf{a}_x dx + \mathbf{a}_y dy + \mathbf{a}_z dz$

Volume Element: $dv = dx dy dz$

Scalar field: $f(x, y, z)$

Vector field: $\mathbf{F}(x, y, z) = F_x \mathbf{a}_x + F_y \mathbf{a}_y + F_z \mathbf{a}_z$

Differential operator ∇ :

$$\nabla = \mathbf{a}_x \frac{\partial}{\partial x} + \mathbf{a}_y \frac{\partial}{\partial y} + \mathbf{a}_z \frac{\partial}{\partial z} = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$$

Gradient: $\nabla f = \frac{\partial f}{\partial x} \mathbf{a}_x + \frac{\partial f}{\partial y} \mathbf{a}_y + \frac{\partial f}{\partial z} \mathbf{a}_z$

Divergence: $\nabla \cdot \mathbf{F}(x, y, z) = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$

Curl: $\nabla \times \mathbf{F} = \text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$

Laplacian: $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$

POLAR AND CYLINDRICAL COORDINATES (ρ, ϕ, z)

Transformation: $x = \rho \cos \phi$, $y = \rho \sin \phi$, $z = z$

Local basis: $\mathbf{a}_\rho = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y$, $\mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y$, $\mathbf{a}_z = \mathbf{a}_z$

Surface element (on $\rho = a$): $d\mathbf{S} = \pm a \mathbf{a}_\rho d\phi dz$

Line Element: $d\mathbf{L} = \mathbf{a}_\rho d\rho + \rho \mathbf{a}_\phi d\phi + \mathbf{a}_z dz$

Scalar field: $f(\rho, \phi, z)$

$$\nabla f = \frac{\partial f}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial f}{\partial \phi} \mathbf{a}_\phi + \frac{\partial f}{\partial z} \mathbf{a}_z$$

$$\nabla \times \mathbf{F} = \frac{1}{\rho} \begin{vmatrix} \mathbf{a}_\rho & \rho \mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ F_\rho & \rho F_\phi & F_z \end{vmatrix}$$

$$\rho = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

Surface element (on $z = \text{const.}$): $d\mathbf{S} = \pm \rho \mathbf{a}_z d\rho d\phi$

Volume element: $dv = \rho d\rho d\phi dz$

Vector field: $\mathbf{F}(\rho, \phi, z) = F_\rho \mathbf{a}_\rho + F_\phi \mathbf{a}_\phi + F_z \mathbf{a}_z$

$$\nabla \cdot \mathbf{F} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho F_\rho) + \frac{1}{\rho} \frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z}$$

$$\nabla^2 f = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial f}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}$$

Solenoid:

$$B = \frac{\mu_0 N I}{L}$$

Toroid:

$$B = \frac{\mu_0 N I}{2\pi R}$$

SPHERICAL COORDINATES (r, θ, ϕ)

Transformation: $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$

Local basis: $\mathbf{a}_r = \sin \theta \cos \phi \mathbf{a}_x + \sin \theta \sin \phi \mathbf{a}_y + \cos \theta \mathbf{a}_z$,

$\mathbf{a}_\theta = \cos \theta \cos \phi \mathbf{a}_x + \cos \theta \sin \phi \mathbf{a}_y - \sin \theta \mathbf{a}_z$,

$\mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y$

Volume element: $dv = r^2 \sin \theta dr d\theta d\phi$

Line Element: $d\mathbf{L} = \mathbf{a}_r dr + r \mathbf{a}_\theta d\theta + r \sin \theta \mathbf{a}_\phi d\phi$

Scalar field: $f(r, \theta, \phi)$

$$\nabla f = \frac{\partial f}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \mathbf{a}_\phi$$

$$\nabla \times \mathbf{F} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{a}_r & r \mathbf{a}_\theta & r \sin \theta \mathbf{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ F_r & r F_\theta & r \sin \theta F_\phi \end{vmatrix}$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

Vector field: $\mathbf{F}(r, \theta, \phi) = F_r \mathbf{a}_r + F_\theta \mathbf{a}_\theta + F_\phi \mathbf{a}_\phi$

$$\nabla \cdot \mathbf{F} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 F_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (F_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi}$$

$$\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}$$

INTEGRATING DERIVATIVES: THE FUNDAMENTAL THEOREM OF CALCULUS (FTC)

Line-integral form:

$$\int_C \nabla g \cdot d\mathbf{L} = \int_C \frac{\partial g}{\partial x} dx + \frac{\partial g}{\partial y} dy + \frac{\partial g}{\partial z} dz = g_{\text{final}} - g_{\text{initial}}$$

Stokes's Theorem:

$$\iint_S (\nabla \times \mathbf{G}) \cdot d\mathbf{S} = \oint_C \mathbf{G} \cdot d\mathbf{L} = \oint_C G_x dx + G_y dy + G_z dz$$

Divergence Theorem:

$$\iiint_V \nabla \cdot \mathbf{G} dv = \iint_S \mathbf{G} \cdot \hat{\mathbf{n}} dS$$

DEFINITE INTEGRALS

$$\begin{array}{lll} \int_0^{\pi/2} \sin x dx = \int_0^{\pi/2} \cos x dx = 1 & \int_0^{\pi/2} \sin^3 x dx = \int_0^{\pi/2} \cos^3 x dx = \frac{2}{3} & \int_0^{\pi/2} \sin^5 x dx = \int_0^{\pi/2} \cos^5 x dx = \frac{8}{15} \\ \int_0^{\pi/2} \sin^2 x dx = \int_0^{\pi/2} \cos^2 x dx = \frac{\pi}{4} & \int_0^{\pi/2} \sin^4 x dx = \int_0^{\pi/2} \cos^4 x dx = \frac{3\pi}{16} & \int_0^{\pi/2} \sin^6 x dx = \int_0^{\pi/2} \cos^6 x dx = \frac{5\pi}{32} \end{array}$$

INDEFINITE INTEGRALS

$$\begin{array}{lll} \int x e^{ax} dx = \frac{e^{ax}}{a^2} (ax - 1) & \int \tan x dx = \ln |\sec x| & \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} \quad (a > 0) \\ \int \sin^2 x dx = \frac{x}{2} - \frac{1}{4} \sin 2x & \int \cos^2 x dx = \frac{x}{2} + \frac{1}{4} \sin 2x & \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) \quad (a > 0) \\ \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) \quad (a > 0) & \int \frac{dx}{(a^2 - x^2)^{3/2}} = \frac{x}{a^2 \sqrt{a^2 - x^2}} & \int \frac{dx}{(x^2 \pm a^2)^{3/2}} = \frac{\pm x}{a^2 \sqrt{x^2 \pm a^2}} \\ \int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) \quad (a > 0) & & \int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln \left| x + \sqrt{x^2 \pm a^2} \right| \end{array}$$

Adapted from R. A. Adams, *Calculus, A Complete Course*, Addison-Wesley, 2003.